

From grading to growth

1 Problem

Large-enrolment modules struggle to provide meaningful feedback on open-ended assessments:

- ⌚ Manual grading can be slow & difficult to scale consistently.
- 💬 Students receive marks or brief annotations without thoroughly understanding their mistakes.
- 🔍 Instructors have limited visibility into cohort-wide misconceptions.
- 💡 Students lack practice targeted to their specific learning gaps.

WHY FEEDBACK MATTERS

94.92%

of students said feedback after a quiz was very important to them — and the majority wanted it in a timely manner, not at term's end.
– Edokpayi, J. N. (2025). *Assessing the impact of frequent quizzes on student performance. Discover Education, 4(480).*

2 Our Solution

AIMS is a human-in-the-loop AI platform that:

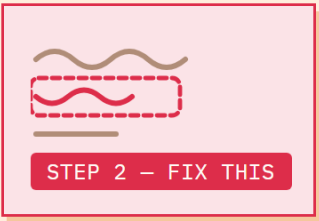
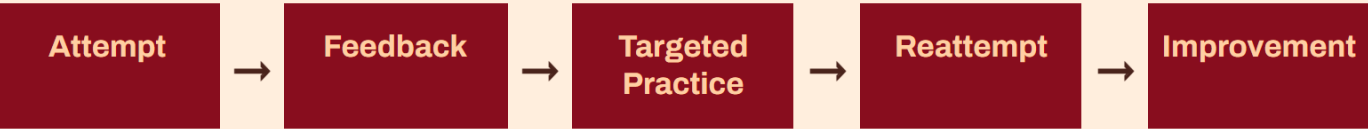
- 👁️ Capable of taking both handwritten and typed/digital responses as input.
- 📊 Suggests rubric-aligned scores with confidence indicators.
- 🎯 Detects misconceptions and valid alternative approaches.
- 💬 Generates personalised feedback and targeted practice.
- 👤 Keeps instructors in control of all grading decisions.

3 Impact

Streamlined, accelerated marking: AIMS takes on the cognitive work of reading each response and mapping it to the rubric, so the instructor's time goes to verifying and assigning the mark — not decoding the handwriting first. The instructor still makes the final decision on every submission.

Personalised feedback at scale: explains what was correct, incomplete, or misunderstood.

Improved mastery: transforms weekly quizzes into a continuous learning cycle.



WHAT STUDENTS GET WITH AIMS

The exact step that went wrong, why it matters, and a practice question to fix it.

Existing platforms vs. AIMS

EXISTING GRADING PLATFORMS	AIMS (OUR UNIQUE SELLING POINTS)
Streamlines rubric-based marking	Supports marking for instructors AND targeted learning for students
Feedback depends largely on manual comments	Automatically drafts misconception-specific feedback
Provides general assignment results	Identifies personalised individual & cohort learning gaps and provides targeted practice questions
Ends after grades are released	Connects mistakes to targeted practice
Rubrics are adjusted as the instructor is marking submissions	Suggests rubric refinements to the instructor before detailed submission review

TOOLS USED	pix2text	AWS Textract	OpenAI API 5.6	RAG	SymPy
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INSTRUCTOR DASHBOARD pictured: the **summative** review screen

► **Guided demo**

sit-markease.vercel.app

photo or scan



pix2text + Textract + RAG + SymPy check + OpenAI 5.6

Formative



?



EXAMPLE HINT

SAFEGUARD

ON REQUEST, FROM EITHER

Summative



W

EXAMPLE FEEDBACK

SAFEGUARD

Student never sees the raw transcription. Low-confidence reads are flagged for visual check.

C

Request a practice revision set

RAG finds material for exactly what was missed; OpenAI turns it into a fresh, verified question that lands straight in the student's practice queue — no instructor gate, since nothing here is graded.